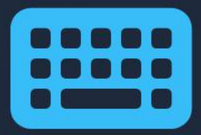


CodeType Trainer

A Real-Time Syntax Typing Tutor Powered by LLMs

Final Project | Jon Steen | Graduate Course CS600

The Problem: Syntax Fluency

- </> **Muscle Memory Gap:** Standard typing tutors use English prose, not code syntax like `{}`, `[]`, or `⇒`.
- </> **Language Specificity:** Developers need to practice the idiomatic patterns of specific languages (Python vs. JS).
- </> **Speed vs. Accuracy:** In live coding interviews or rapid development, syntax errors break flow.
- </> **Goal:** Build a tool that generates infinite, valid coding scenarios for practice.

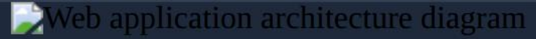
The Solution: CodeType Trainer

AI-Generated Content

Leverages a fine-tuned GPT-3.5 model to generate randomized, syntactically correct code snippets on demand. No static databases.

Real-Time Metrics

Tracks WPM, accuracy, and error count character-by-character using a custom JavaScript implementation embedded in Streamlit.



System Architecture

- **Frontend:** Streamlit (Python) handles state management and UI layout.
- **Backend Logic:** OpenAI API (`ft:gpt-3.5-turbo`) generates content dynamically.
- **Bridge:** `streamlit.components.v1` renders a custom HTML/JS isolated iframe.
- **Data Flow:** Python fetches snippet → Sends to JS → JS captures events → Returns metrics to Python.

Technology Stack

Python & Streamlit

Core application logic, session state management (`st.session_state`), and UI framework.

OpenAI API

Fine-tuned model `ft:gpt-3.5-turbo-0125` for strict output formatting and difficulty scaling.

HTML5 / JS

Custom component for low-latency event handling (keydown listeners) and visual feedback.

The Fine-Tuned Model

Why fine-tune `gpt-3.5-turbo`?

- </> **Strict Formatting:** The model is trained to return raw code *only* (no markdown, no chatty explanations).
- </> **Difficulty Tiers:** Custom logic for `BEGINNER`, `INTERMEDIATE`, and `ADVANCED` complexity.
- </> **Negative Constraints:** Trained to avoid repeating patterns from previous snippets context.
- </> **Efficiency:** Smaller prompts and faster completion times compared to few-shot prompting.

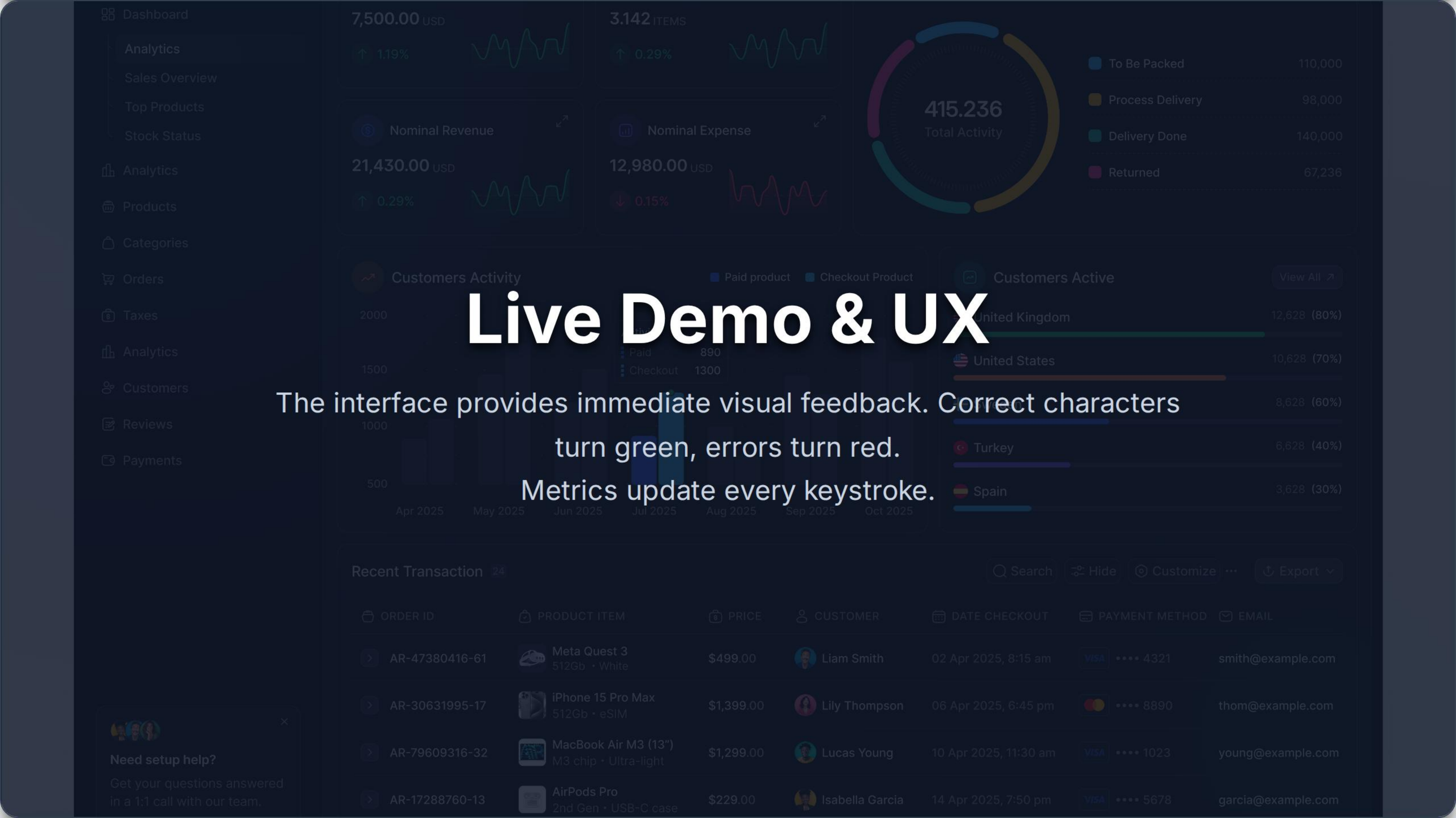
Key Implementation Details

State Management

Used `st.session_state` to persist the current snippet, user stats, and history across re-runs. This prevents the snippet from refreshing whenever a user interacts with a widget.

The Custom Component

Streamlit's default text area doesn't support real-time color highlighting. I injected a custom HTML/CSS block using `components.html` to create a reactive typing surface that calculates WPM in the client browser.



Live Demo & UX

The interface provides immediate visual feedback. Correct characters turn green, errors turn red.
Metrics update every keystroke.

Development Challenges

Prompt Leakage

Initially, the model would wrap code in markdown blocks. Fine-tuning datasets had to be scrubbed to enforce "raw code only" output.

Latency

Generating code on the fly takes time. Solution: Caching strategies and optimized token limits for the API response.

Component Sync

Passing data from the JS sandbox back to Python Streamlit state required careful URL parameter handling.

Future Improvements

1

Auth

User accounts to save long-term progress history.

2

Languages

Add support for Rust, TypeScript, and SQL.

3

Leaderboard

Global ranking system for competitive typing.

4

IDE Plugin

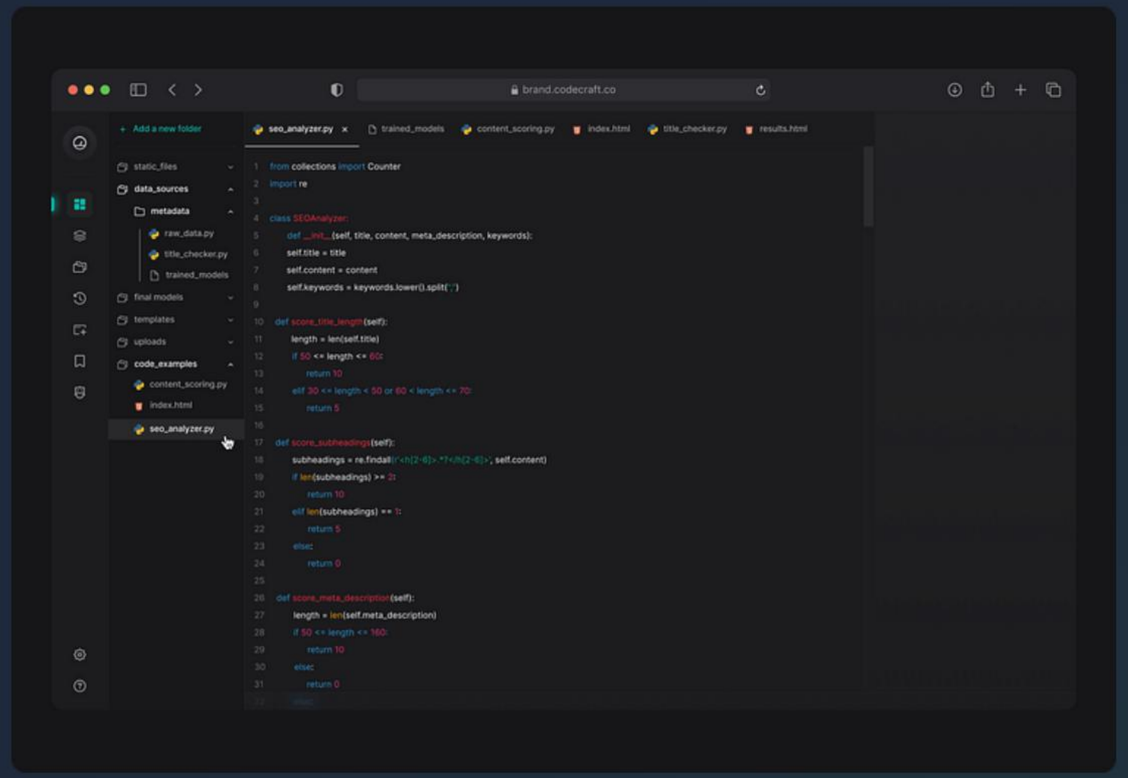
Port logic to a VS Code extension.

Project Summary

CodeType Trainer

A full-stack application demonstrating the power of integrating Generative AI with traditional web frontend logic.

- </> **Functional:** Real-time syntax practice.
- </> **Technical:** Advanced usage of Streamlit Components.
- </> **AI-Driven:** Custom Fine-Tuned Model.



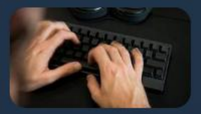
Questions?

Thank you for your time.

/jonsteen

jon@example.com

Image Sources



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